

Evaluating Convolutional Neural Network Performance on MRI Scans and Low-Saliency Shape Recognition

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Abstract—This research paper aims to explore performance and the underlying reasons behind MRI scans and pictures with weak edge definition. MRI scans are often not difficult to classify from an extreme binary perspective (Non Dementia vs Extreme Dementia), but from a scale, CNN models often struggle to classify certain images into their correct category. Overall, we found that maintaining the spatial integrity of the images, without flattening them, causes the greatest effects on precision and recall of these classifications. To further improve these classifications-recall, especially, it is imperative that the model is able to reweight channels dynamically, allowing it to extract useful signal even from images where edge metrics are uninformative.

Index Terms—DementiaCNN, Le-Net 5, MLP, UMAP, Canny, Sobel, F1-Score

I. INTRODUCTION (JOSHUA: 90% GERARDO: 10%)

Dementia is a progressive, neurological disorder, which is a major health burden in rural areas with limited access to doctors and diagnostic equipment. [1] The observed dementia prevalence is higher in rural areas than in urban areas, which is largely due to the structural disadvantages rural residents have over urban residents, including lower levels of education, more multi-chronic conditions, and less access to quality health care services. In addition, there is further evidence that older age people in these areas may be underdiagnosed or diagnosed at a later stage of dementia than in urban settings [2], highlighting the need for readily accessible screening tools. This project is motivated by the opportunity to obtain a low cost, readily available hardware like phones, to be able to process them with automated systems that can be used to classify dementia from MRI scans, which is otherwise dependent on expensive clinical infrastructure in these poorer regions.

A key challenge in this research is the development of accurate classifiers for brain MRI data that can be automated. MRI imaging is a common method used to detect

structural brain changes associated with the progression of dementia without requiring any invasive procedures, and has been shown to provide a non-invasive means to visualise atrophy and other neural changes during dementia phases [3]. When researchers scan dementia patients, though, they find it challenging to classify the differences, as the visual differences between the cases - mild, very mild, moderate, and non-demented - are subtle and hard to isolate into distinct visual features. This is an example of what we call low-saliency shape recognition - distinguishing features between classes are not very distinct or sharp.

The goal of this project is to understand how a Convolutional Neural Network can accurately detect features in images that are corrupted or have low visual saliency, with the specific application of classification of dementia MR scans. We trained a LeNet-5, MLP and our own custom architecture on a labeled dataset of MRIs from patients with dementias to systematically assess the behavior of the models. LeNet-5, a basic CNN network introduced in the paper Gradient-Based Learning Applied to Document Recognition [4], is a controllable linear baseline that is interpretable in terms of the architecture of CNN layers. We used the Canny and Sobel edge detection operators to compute quantitative edge sharpness and shape irregularity of every image, and compared the quantitative measures with the classification results of the model. To demonstrate the effect of data quality and class structure on the results, we also did a parallel evaluation using Fashion-MNIST [5], a well studied reference data set of clean high saliency grayscale clothing images where the class distribution is balanced, thus separating out the effect of data quality and class structure from the effect of model architecture.

Results from the edge sharpness and irregularity seemed to suggest that image distortion was compulsory for creating an accurate classification model, but dynamic

feature learning became more important for improving an already decent model. In addition, an imbalance in class distribution, in favor of the majority (non demented), was observed to have inflated the overall accuracy as clinically critical minority classes performed much lower. The results have applications for the use of lightweight CNN architectures in the clinic and suggest class rebalancing and more complex feature learning as the main paths forward.

The rest of this report details the datasets and preprocessing, the Convolutional Neural Network architecture and training method, the methodology for analyzing the edges and a detailed discussion of the results and implications for accessible screening for dementia.

II. RELATED WORK(JOSHUA: 10%, GERARDO:90%)

In the LeNet paper [4], the authors present a solution to the drawbacks of fully connected networks: LeNet 5. One of the oldest and most impactful architectures of convolutional neural networks (CNNs). The biggest drawback to fully connected networks was that they considered each pixel on the image independently, LeNet addressed this by introducing convolutional layers, which have local receptive fields and weight sharing. Each neuron in the system can only examine a small portion of the image, because of their local receptive fields. Weight sharing is the same filter used across the whole image, thus so that the network can learn the features anywhere. This solution enables automatic learning of spatial features in the network. This architecture was the foundation of convolution, pooling and fully connected layers, which are the fundamental principles of CNN design that remain true to this day. LeNet 5 was developed for the simple and grayscale image database called MNIST. The MNIST dataset consists of 70 thousand grayscale 28x28 image dataset of 10 different classes, each having images of one of the numbers from 0-9. LeNet was able to demonstrate CNNs could outperform fully connected networks on image classification with its strong results on the MNIST dataset. The architecture was not designed for medical and complex images, and questions have been raised as to how well the architecture will scale to more complex images, like MRIs. These questions are in the scope of this project. The Fashion MNIST paper [5] reasoned that MNIST was too trivial to provide a meaningful test for current models. It presented Fashion MNIST with 10 fashion classes as opposed to numerical digits. The data format was identical to the original MNIST dataset, with 70 thousand images of 28x28 pixels in a gray scale. This was done to enable scientists to use it on their existing pipelines. Even the most sophisticated models were only getting 90 percent accuracy, which is a big drop from the 99 percent accuracy achieved on the MNIST data. This project uses Fashion MNIST as the other dataset to be able to evaluate the MLP and LeNet 5 models under more realistic and challenging conditions.

There is a paper by Kshatri and Singh [6] that provided a survey of CNN applications in medical image analysis particularly in MRI. Convolutional Networks (CNN) are the most prevalent approach in medical image analysis since 2012. The main two problems in using CNNs for medical image analysis are the limited availability of labeled training data and subjective labeling. Unlike other categories, medical imaging can only be labelled by medical practitioners such as radiologists and doctors. This is both expensive and time consuming, causing medical datasets to generally be much smaller than general image datasets. Subjective labelling may occur because the same image is labelled differently by different health care workers. The medical classification does not have any objective ground truths, they are based on opinions of professionals. This project is based on the MRI dataset where folders are implicit labels. The labels were given by the designer of the database, and might have been given in a similar way as was discussed earlier. This project uses grayscale conversion, resizing the MRI to 32x32 pixels, and scaling the pixel values to the [0, 1] range to ensure the data are accurate in the MRI preprocessing pipeline, which may directly skew the classification accuracy. This paper [6] goes over the importance of edge preservation during preprocessing. The purpose of this project is to quantify the ability to preserve edges while measuring the relationship between edge preservation and accuracy, further testing previous claim. Although the paper mentions transfer learning and data augmentation as methods to overcome limited data, it can be seen that the MRI dataset in this project comprises more than 86 thousand images. In this case, that number is enough and unnecessary to make use of those techniques. This project will use accuracy as well but also incorporate the use of confusion matrices, classification reports and edge irregularity analysis, which were not used in the approaches reviewed in the paper. The Lin et al. paper [7] applies CNNs to brain MRIs to classify stages of Alzheimer's disease, very similar to what this project will do. The experiment demonstrated that convolutional feature extraction is effective enough from the MRI to be able to differentiate among the various stages of neurodegeneration. In this paper, the classification accuracy is satisfactory, indicating that CNNs can be used to classify the stage of dementia. A limitation was it only focused on classification accuracy as a measurement. The aim of this project is to extend this work by defining 4 stages of the dementia and to introduce the dimension of edge irregularity as an additional one to analyze and to compare both LeNet and MLP on the same set.

In Bhalla et al. [8], they have come up with a method for multi-focus image fusion using fuzzy sets and CNN. The uncertainty and noise are processed by fuzzy sets, and CNN is used for feature extraction and focus of regions. Multi-focus image fusion is a technique that merges multiple images at different focus lengths into a single fully-focused image. The main result of this paper was that the edge information is a very important criterion

for measuring image quality. Poorly defined edge and uncertain edge images resulted in poor model output and sharp and well defined edge images resulted in good fusion quality and good classification performance. They used a measure of the transfer of visual and boundary information from source images to the output, called EI. The result was that the higher the EI values the better the quality outputs with clearer boundaries, comparing the EI values across 7 different methods. The Canny, Sobel and irregularity analysis in this project also quantify the relationship between the complexity of edges and model performance [8]. This work is an extension of their work on image fusion to image classification. This project extends the edge irregularity metric by being able to take into account the directionality of the edge as well as the density, which means that it can detect wrinkle-like patterns that simple edge score metrics could not detect.

III. OUR APPROACH (JOSHUA: 60%, GERARDO: 40%)

In our experiment we made use of 3 models: LeNet-5, MLP and custom neural network design to deal with grayscale MRI scans. Of the pre-built models, we placed greater emphasis on results from LeNet 5 due to the higher sophistication when it came to feature extraction and preservation of spatial structure.

A. LeNet 5

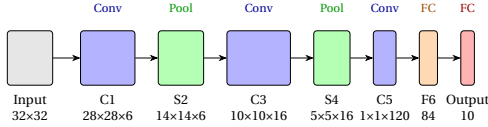


Fig. 1. LeNet-5 Architecture

LeNet-5 is a CNN composed of two convolutional layers followed by ReLU activation and 2x2 max-pooling layers and three fully connected layers leading to a reduction in the number of output classes.

The architecture was designed in a hierarchical fashion, with a focus on spatial feature extraction. The convolutional layers with 5x5 filters gradually extract local features like edges and textures, while the pooling layers create spatial invariance and dimensionality reduction. The two convolutional stages have 6 and 16 feature maps as output, while the flattened output is fed to fully connected layers of sizes 120, 84 and finally the number of target classes. After training for 10 epochs and using the Adam optimizer with a learning rate of 0.001.

$$L = - \sum_{i=1}^n y_i \log(\hat{y}_i)$$

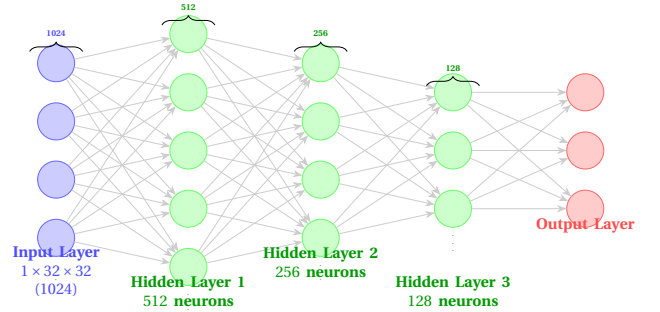
The model is trained with a categorical cross-entropy loss. Since this is a four-class (or even 10-class, for Fashion-MNIST) classification problem, this loss is computed as the negative log-likelihood of the class that is

predicted, and will punish when a class is confidently mis-predicted, which is appropriate for this kind of classification problem with a softmax-type output.

LeNet-5' has the main benefit of learning locally distributed image features by sharing the same convolutional weights across space, reducing the number of parameters needed in comparison to fully-connected approaches, while still being sensitive to structural features in images. The progressive pooling stages also have translation invariance. What is relatively shallow is the architecture, however, that means it has limited ability to model complex and highly abstract features. The small 5x5 kernel size might also fail to pick up on wider contexts that cover larger areas of the image.

We made a couple of changes compared to the traditional LeNet-5 formulation. All ReLU activations have been used to replace the original sigmoid and tanh functions throughout, which have a better gradient flow and accelerating convergence. Average pooling was replaced with max-pooling, which maintains the strongest activations and thus is more likely to create strong representations of features. The output layer was also modified to have four classes for the task of dementia classification instead of the original ten-class digit output and ten classes for Fashion-MNIST.

B. Multi-layer Perceptron



The MLP is made of flattening layer followed by three fully connected hidden layers with size 512, 256 and 128 neurons, and use ReLU as activation function, with dropout layers after the first two layers and finally a linear output layer of size four.

The network design was chosen to be a simple fully connected baseline. The images are first flattened and converted into a 1024 dimensional vector and are fed into a series of increasingly narrow hidden layers. The first two ReLU activations are dropped with a rate of 0.3 as a regularizer during training and the Adam optimizer is employed with the same learning rate and number of epochs as LeNet-5. Supervision is by the same cross-entropy loss function as above.

One of the best things about the MLP is that it can be implemented and trained quickly and easily, and it can be used as a benchmark for comparison. The chief drawback is that it has no spatial awareness. The model discards all positional and structural relationships between pixels because it just flattens the image prior to processing. This

renders it more difficult to adapt to image classification compared to convolutional methods, where correlations between raw pixels would be learnt over the entire input dimension. The architecture also includes a lot of parameters at the first layer (1024 to 512), that without the regularization afforded by the dropout layers, could easily overfit.

The MLP architecture in our case, was chosen due to its simplicity. It will serve as a "Straw Man" model, where significant differences in results between the other two complex models will likely suggest that our dataset has enough spatial features to warrant a CNN.

The most important change made by the team was to add dropout regularization after the first two hidden layers, which is not used in a standard MLP formulation and helps the network generalize better on a test set.

C. Custom Architecture

TABLE I
DEMENTIACNN ARCHITECTURE FOR MRI DEMENTIA CLASSIFICATION

Block	Layer	Output Shape	Details
	—	$(B, 1, 32, 32)$	Grayscale MRI patch
Input	Conv2d	$(B, 32, 32, 32)$	3×3 , pad=1, BN, ReLU
Block 1	Conv2d	$(B, 32, 32, 32)$	3×3 , pad=1, BN, ReLU
	MaxPool2d	$(B, 32, 16, 16)$	2×2
	Squeeze-Excite	$(B, 32, 16, 16)$	Reduction=8
Block 2	Conv2d	$(B, 64, 16, 16)$	3×3 , pad=1, BN, ReLU
	Conv2d	$(B, 64, 16, 16)$	3×3 , pad=1, BN, ReLU
	MaxPool2d	$(B, 64, 8, 8)$	2×2
	Squeeze-Excite	$(B, 32, 16, 16)$	Reduction=8
Block 3	Conv2d	$(B, 128, 8, 8)$	3×3 , pad=1, BN, ReLU
	MaxPool2d	$(B, 128, 4, 4)$	2×2
	Squeeze-Excite	$(B, 128, 4, 4)$	Reduction=8
DS Conv	Depthwise Conv	$(B, 128, 4, 4)$	3×3 , groups=128
	Pointwise Conv	$(B, 128, 4, 4)$	1×1 , BN, ReLU
Head	Avg Pool	$(B, 128)$	Global average pooling
	Dropout	$(B, 128)$	$p = 0.4$
	Linear	$(B, 256)$	ReLU
	Dropout	$(B, 256)$	$p = 0.2$
	Linear	$(B, 4)$	Logits
Output	—	$(B, 4)$	One logit per dementia class

The 32×32 single-channel MRI patches are fed into three successive deeper convolutional blocks, a depthwise-separable refinement stage, and a lightweight classifier head, resulting in the creation of DementiaCNN. Each block is reduced in size by half, from 32×32 to 16×16 , 8×8 , 4×4 , in accordance with the usual principle that sacrificing spatial resolution in exchange for representational richness, channel depth is increased from 32 to 64 and 128. The 4×4 feature maps are reduced to a single vector of 128 dimensions that is fed to the fully connected head, which generates four class logits, one for each dementia stage, in the results.

Two convolutions per Block. Blocks 1 and 2 each have two 3×3 convolutions followed by a pooling layer. The single convolution layer shown in each of the scales is enough to cover simple texture discrimination, however, for dementia staging to be sensitive, subtle differences

in the structure, such as slight cortical thinning, early atrophy of the hippocampus and mild increase in the size of the ventricles that appear only after two non-linear passes, covering richer local representations, are discarded by pooling. It's like the philosophy of VGG-style stacking, that is, depth before downsampling is better for gradient signals in order to enable more complex interactions between features, but not at the expense of larger kernel parameters. Squeeze-and-Excite (SE) attention. Each block contains an SE module that applies global average pooling over the spatial dimensions and feeds a small 2-layer MLP followed by an element-wise multiplication with the feature maps. This in practice enables the network to learn which feature maps are diagnostically important and to ignore the rest. Not every 32 or 64 feature maps will have important anatomy, in the case of MRI dementia classification, some will pick up on scanner noise or artefacts, and the SE block is a learned, data-driven way to down-weight those channels so they don't propagate further. At 32×32 resolution this is especially useful because there is not much spatial context to disambiguate useful structure from noise. Depthwise-separable convolution. After Block 3, a depthwise-separable layer applies a 3×3 filter separately to each of the 128 channels (depthwise step) and then combines them with another 1×1 pointwise convolution. This factorisation allows to cut down the number of parameters and speed up the computation compared to a standard 3×3 Conv2d, but still keeps the same receptive field. It is most of all a fine-detail refinement stage at the smallest spatial scale (4×4), in which the network is aiming at either a coarse- or fine-grain abstraction of the most discriminative detail prior to the elimination of all spatial structure in the global pooling stage. A complete convolution would be parameter-intensive given the information available; the separable variant has the same functional goal less parameter intensive. Global Average Pooling + Classifier head. DementiaCNN does not flatten the 4×4 feature maps as LeNet-5 does, but rather uses global average pooling first. This creates a 128 dimensional summary vector, irrespective of the input resolution, thus dramatically reducing the number of parameters in the head, and serving as a natural regulariser. The head is composed of two linear layers, 128 to 256 and 256 to 4, preceded by dropout rates of 0.4 and 0.2 respectively, giving a graduated regularisation, strongest at the most abstract level of representation.

DementiaCNN is trained using cross-entropy loss, which is the most common loss function for multi-class classification, similar to LeNet-5 and the baseline MLP. Here the use of cross-entropy is suitable since it directly optimises the log-probability of the right class, giving well-scaled gradients during training. The network is small label smoothed with 0.05 to guard against overconfidence on the training classes (which is a significant worry when the visual distinction between the Non Demented and Very Mild Dementia is in fact not clear even to clinicians).

Advantages. One of the most important advantages over

LeNet-5 is its ability to learn multiple scales at once: DementiaCNN. The design of LeNet-5 was for 32×32 digit images in which the features of interest (stroke width, closed loops, endpoints) are relatively coarse. The dementia staging from MRI imaging relies on much subtler cues, spread out across spatial scales: global atrophy patterns, at the coarsest; local textural changes, at the finest. The dual-convolution blocks with SE attention enable the network to construct increasingly abstract representations and suppress uninformative feature maps at each scale, while LeNet-5’s fixed, shallow architecture cannot do. In addition, the SE attention mechanism enhances clinical interpretability of the network. Upon inspection, it is in principle possible to find out which feature maps the SE layer amplifies, thus giving a soft indication of the characteristics of the image that the model finds diagnostically relevant. The other benefit is the depthwise-separable stage which allows the spatial filtering to be performed independently of the channel mixing, which lowers the risk of overfitting at the lowest spatial scale on which the feature maps are most abstract and the training signal is weakest. Last but not least, the global average pooling approach is much more parameter efficient than flattening approach adopted by LeNet-5. This is important since the classifier head is very often the place where overfitting occurs in small CNNs — the fewer parameters the less likely the model is to overfit on training examples and hence memorise them. Disadvantages. The additional complexity compared to LeNet-5 is a real cost. There are more components that interact with each other in DementiaCNN: SE bottlenecks, depthwise separable layers, graduated dropout, and this makes hyperparameter tuning more difficult. Able to diagnose which of the other three (suboptimal choices for the SE reduction ratio, dropout rates, and the depthwise kernel size) is causing performance degradation, but it is harder to do with a more complex model than a simpler model. The other basic restriction is the input resolution. All three models are based on 32×32 patches and thus lose much of the spatial information in a clinical MRI scan. These subtle structural differences between Very Mild Dementia and Mild Dementia, which a radiologist is able to see at full resolution, may just not be recoverable, no matter the architecture. DementiaCNN’s additional capacity is not enough to make up for the information that has been lost during pre-processing. In this respect, the architecture is more expressive than it is strictly needed by its inputs, and the ceiling is defined by resolution and not by architecture expressiveness. Overfitting can also happen with small data sets. The SE modules and the two-convolution blocks add more parameters than LeNet-5, but global average pooling and dropout remove these parameters, requiring careful tuning of the regularisation budget. If the dataset is small, the simpler LeNet-5 model can perform well, or even better, not because it’s a better model, but because its smaller capacity is well suited to the amount of information in the dataset.

The idea behind the motivating hypothesis of DementiaCNN is that there is useful discriminative signal at multiple scales and in multiple channels, and there’s a model that can selectively focus on the most relevant signal at each scale and in each channel that will outperform a model with fixed filters applied non-discriminatively. It isn’t designed with this in mind, LeNet-5 was designed to fit the requirements of digit recognition from the 90’s, and it does not have any mechanisms that dynamically modulate feature importance. The MLP baseline is the worst, as it ignores any spatial structure, showing that spatial relationships have signal value. Therefore, one should expect to see DementiaCNN perform better on classes for which the distinguishing features are multi-scale, or where there is noise, such as between adjacent classes like Very Mild Dementia and Non Demented. The advantage in performance might be modest in the more clearly distinguished cases, e.g., Non Demented vs fully Demented, on which even a superficial network can learn those coarser discriminations. The true test of the architecture will be whether the SE attention and dual-convolution depth can produce enough additional representational power to capture the subtle, continuous progression of cognitive decline, which does not have a clear cut division into the four distinct visual categories.

IV. EMPIRICAL SETTING (JOSHUA: 40%, GERARDO: 60%)

A. MRI Dataset description

This study employs a MRI brain scan database which is available in a single zip file. Instead of giving explicit class labels in a metadata file or annotation scheme, the archive is organised into four subdirectories, whose names explicitly encode the diagnostic category of each of the images contained in them: Mild Dementia, Moderate Dementia, Non Demented, and Very Mild Dementia. This is an implicit labelling approach: each image is sorted into one of these label folders and they are sorted into a csv file as such, assigning to each folder a label number 0, 1, 2 or 3. The data set then covers four classes which represent four clinically relevant stages of cognitive decline. All images have been resized to 32×32 , which results in a 1,024 element flat feature vector for each image before any additional transformations are applied. The entire data set consists of around 86,000 images. These were divided into a 70 / 15 / 15 split into training, validation and test sets, leaving 60,505 training images, 12,965 validation images and 12,967 test images. The splitting was performed with random sampling without stratification and all three subsets have the same image resolution and processing procedure. Because the split was done by image rather than by patient, neighboring slices from the same scan can appear in both the training and test sets.

B. Pre-processing

Images were fed through the torchvision pipeline in PyTorch, and then all images were pre-processed in the same way: grayscale (if not already single channel), resized

to 32×32 pixels and converted to floating-point tensors. No data augmentation was used in the training. The preprocessed images were flattened from a 32×32 spatial structure to 1024 dimensions to perform dimensionality reduction and subsequently standardized using z-score normalization (standardizing each feature by subtracting the mean and dividing by the standard deviation of the feature) with the help of StandardScaler for downstream visualization. PCA is then performed on this dataset, in order to reduce noise and overfitting, leaving about 95 percent variance. This PCA-reduced representation was then used as the input for the UMAP stage to speed up data visualization which is discussed later.

C. Model Architectures and Hyperparameters

Three neural architectures were trained and evaluated: a LeNet-5 convolutional network, a multilayer perceptron (MLP), and a custom CNN (DementiaCNN) incorporating squeeze-and-excitation attention and depthwise-separable convolutions.

LeNet-5 and MLP. The models were trained with the Adam optimizer with a learning rate of 0.001 and a batch size of 32. The training was performed over 10 epochs using cross-entropy loss. Adam was chosen for its adaptive per-parameter learning rates and proven convergence properties for image classification problems.

Both Adam and AdamW build upon Momentum(m_t) and RMSProp(v_t) which are used to accelerate the gradient descent and overcomes the problem of diminishing learning rates

$$\begin{aligned} m_t &= \beta_1 m_{t-1} + (1 - \beta_1) \nabla f_t(\theta_{t-1}) \\ v_t &= \beta_2 v_{t-1} + (1 - \beta_2) (\nabla f_t(\theta_{t-1}))^2 \end{aligned} \quad (1)$$

(First and second moment updates)

where:

- m_t is the moving average of the gradient at time t
- v_t is the exponentially weighted average of squared gradients
- β_1 is the momentum parameter
- β_2 is the RMS parameter

D. UMAP

Uniform Manifold Approximation and Projection(UMAP), is a gold standard dimension reduction technique based on Riemannian geometry and algebraic topology theory. Because of its great visualization and computing performance it is widely used in the biomedical sphere.

The algorithm below, n is the number of neighbors of the local metric; d is the target embedding dimension; min-dist is the distance or tightness for the embeddings and $n\text{-epochs}$ is the number of training iterations.

Similar to t-SNE, UMAP can be divided into two stages: Building a High Dimensional data graph represented as Fuzzy Topology. This captures the local connectivity and

Algorithm 1: Uniform Manifold Approximation and Projection

```

1: Function  $UMAP(X, n, d, \text{min-dist}, n\text{-epochs})$  is
2:   forall  $x \in X$  do
3:      $\text{fs-set}[x] \leftarrow \text{LocalFuzzySimplicialSet}(X, x, n)$ 
4:    $\text{top-rep} \leftarrow \bigcup x \in X \text{fs-set}[x]$ 
5:    $Y \leftarrow \text{SpectralEmbedding}(\text{top-rep}, d)$ 
6:    $Y \leftarrow \text{OptimizeEmbedding}(\text{top-rep}, Y, \text{min-dist}, n\text{-epochs})$ 
7:   return  $Y$ 

```

topological structure of the data points. It involves calculating the n -neighbors(nearest neighbor) for each data point.

Secondly is to calculate the low-dimensional layout of the graph. The algorithm iteratively applies repulsive(RFE) and gravitational(GFE) forces at each vertex or edge.

UMAP was used instead of t-SNE for data visualization in this case due to its superior ability to handle large datasets and preserve global organization of data, making it's clusters tighter. We set n -neighbors to 30 for increased specification, for better clustering of data

E. Evaluation Metrics

To evaluate model performance, classification accuracy on the test set and per-class precision, recall and F1-scores through complete classification reports were used. Confusion matrices were also calculated to investigate misclassification patterns on the four severity classes of dementia..

In addition to standard classification metrics, edge-based image quality analyses were conducted using **Canny edge detection** and **Sobel gradient filtering** to find the relationship between image sharpness and model accuracy.

Canny edge scores capture the density of detected edges in an image, while Sobel-based scores measure gradient magnitude and an edge irregularity index derived from gradient direction variance. These metrics were computed per test image and binned to produce accuracy-vs.-sharpness curves, allowing examination of whether images with stronger or more complex edge structure were associated with higher or lower classification performance. Visualizations of these analyses are presented in the Results section.

Sobel(Kernel Convolutional Process) gradient filtering developed by Irwin Sobel and Gary Feldman, are held by **two** operating kernels that measure changes in intensity in the **horizontal** and **vertical** directions.

$$G_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix} * A$$

$$G_y = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix} * A$$

Horizontal Kernel (G_x) and Vertical Kernel (G_y) are placed over each pixel and are multiplied by Image (A) and the corresponding values are added.

$$G = \sqrt{G_x^2 + G_y^2}$$

$$\Theta = \text{atan2}(G_y, G_x)$$

Magnitude defined through G as it simply uses the Pythagorean Theorem of the horizontal and vertical operators. Large numbers of G indicate that an edge has been found.

Canny is an improved version of Sobel. It takes the images from the Sobel output and thins the edges so they are one pixel wide. Problem with Sobel is that when you find the edges, you still see gradients in these edges that creates a lot noise and makes it messy, canny simplifies this. Gaussian smoothing($K(x, y)$) is implemented in order to do this.

$$K(x, y) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{x^2 + y^2}{2\sigma^2}\right) \quad (1)$$

Firstly Canny determines if the pixel is bigger than it's neighbors, bearing in mind the **orientation** of the edge. This stage thins the edges to remove gradient

$$M(x, y) = \begin{cases} G(x, y) & \text{if } G(x, y) > G(x_{neighbors}) \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

It then applies hysteresis thresholding, which -determined by the researcher- keeps edges of a certain strength typically from the scale of 0 to 255, the rest are thrown away

$$\text{Edge}(x, y) = \begin{cases} \text{Strong} & \text{if } G \geq T_{high} \\ \text{Weak} & \text{if } T_{low} \leq G < T_{high} \\ \text{None} & \text{if } G < T_{low} \end{cases} \quad (3)$$

The result images of both Canny and Sobel are grayscale images of basically the outlines of the input image. We utilized this because MRI scans of dementia for patients in their early stages can be very unpronounced. The idea was to see if higher accuracies had a relationship with higher edge sharpness and irregularity scores, which could be and indication of overfitting/underfitting by the models.

A. Dataset Overview

The Dementia dataset is made up of 86,437 MRI images across 4 classes. This includes 67,222 images in the Non Demented class, 13,725 in the Very Mild Dementia class, 5,002 in the Mild Dementia class, and 488 in the Moderate Dementia class. 70% or 60,505 of the images will be used for training, while 12,965 images (15%) are used for validation and 12,967 (15%) for testing. The grayscale images were resized to 32x32 pixels to ensure consistency. Two notable limitations of this dataset are that Non Demented makes up 77% of the dataset while Moderate Dementia is underrepresented at only 1%. The Fashion MNIST dataset has 70,000 images across 10 classes. The 10 classes are T-shirt, Trouser, Pullover, Dress, Coat, Sandal, Shirt, Sneaker, Bag, and Ankle Boot. It comes with 60,000 training images and 10,000 testing images, but an 80-20 split is still applied to the training set. The grayscale images were resized from 28x28 to 32x32 to match the dementia MRI dataset. This dataset is very balanced across the 10 classes compared to the dementia MRI dataset. The LeNet 5 and MLP models will be evaluated on both datasets while our custom made DementiaCNN will be evaluated on the Dementia dataset only.

B. Dementia Dataset Results

LeNet, MLP, and DementiaCNN were evaluated on the Dementia dataset. LeNet had a 98.96% test accuracy, MLP had a 93.57% test accuracy, and DementiaCNN had a 99.99% test accuracy.

Epoch 01/10	Train Loss: 0.5382	Acc: 79.52%	Val Loss: 0.4623	Acc: 82.75%
Epoch 02/10	Train Loss: 0.3271	Acc: 87.15%	Val Loss: 0.2560	Acc: 90.01%
Epoch 03/10	Train Loss: 0.1628	Acc: 93.90%	Val Loss: 0.2137	Acc: 91.50%
Epoch 04/10	Train Loss: 0.0822	Acc: 97.06%	Val Loss: 0.0516	Acc: 98.29%
Epoch 05/10	Train Loss: 0.0537	Acc: 98.08%	Val Loss: 0.0541	Acc: 98.06%
Epoch 06/10	Train Loss: 0.0369	Acc: 98.72%	Val Loss: 0.0307	Acc: 98.97%
Epoch 07/10	Train Loss: 0.0317	Acc: 98.87%	Val Loss: 0.0239	Acc: 99.19%
Epoch 08/10	Train Loss: 0.0279	Acc: 99.05%	Val Loss: 0.0387	Acc: 98.74%
Epoch 09/10	Train Loss: 0.0241	Acc: 99.18%	Val Loss: 0.0794	Acc: 97.20%
Epoch 10/10	Train Loss: 0.0197	Acc: 99.32%	Val Loss: 0.0315	Acc: 98.94%

LeNet-5 Dementia test accuracy: 98.96%

Fig. 2. Results from LeNet model being applied to the Dementia dataset

As illustrated in Fig. 2, LeNet 5 had an impressive test accuracy of 98.96%. The training accuracy was at 99.32% and validation accuracy at 98.94% after epoch 10. The loss began at .5382 and ended at .0197, constantly decreasing throughout the epochs. From epoch 4 onward the training and validation accuracies tracked within about 2% of each other, apart from a brief validation dip to 97.20% at epoch 9 that recovered right away. Therefore there were no signs of overfitting. There was also strong and consistent convergence across all 10 epochs. LeNet was trained using the Adam optimizer with a learning rate of 0.001 over 10 epochs.

As shown in Fig. 3, MLP had the lowest test accuracy at 93.57%. The training accuracy was at 91.11% and validation accuracy at 93.55% after epoch 10. The loss began at .5420 and ended at .2198, showing a slower decrease than in LeNet. The validation accuracy was consistently a

Epoch 01/10	Train Loss: 0.5420	Acc: 79.31%	Val Loss: 0.4638	Acc: 80.86%
Epoch 02/10	Train Loss: 0.4331	Acc: 82.32%	Val Loss: 0.3552	Acc: 85.58%
Epoch 03/10	Train Loss: 0.3748	Acc: 84.57%	Val Loss: 0.3138	Acc: 86.78%
Epoch 04/10	Train Loss: 0.3373	Acc: 86.89%	Val Loss: 0.3001	Acc: 87.02%
Epoch 05/10	Train Loss: 0.3051	Acc: 86.91%	Val Loss: 0.2252	Acc: 91.32%
Epoch 06/10	Train Loss: 0.2875	Acc: 87.70%	Val Loss: 0.2211	Acc: 91.49%
Epoch 07/10	Train Loss: 0.2597	Acc: 89.22%	Val Loss: 0.1794	Acc: 93.35%
Epoch 08/10	Train Loss: 0.2572	Acc: 89.12%	Val Loss: 0.1956	Acc: 93.04%
Epoch 09/10	Train Loss: 0.2434	Acc: 89.91%	Val Loss: 0.1540	Acc: 94.69%
Epoch 10/10	Train Loss: 0.2198	Acc: 91.11%	Val Loss: 0.1807	Acc: 93.55%

MLP Dementia test accuracy: 93.57%

Fig. 3. Results from the MLP model being applied to the Dementia dataset

few points higher than the training accuracy, which is an unusual pattern. This is most likely caused by dropout. The 0.3 dropout layers are active while training accuracy is measured but switched off during validation. Like the other two models, the MLP was trained on the raw 32x32 pixels, flattened to a 1024-value vector, and it still reached almost 94%, showing that raw pixel intensities carry a lot of signal. The 5% difference between LeNet and MLP does confirm that spatial feature extraction adds value.

Epoch 01/25	Train Loss: 0.4655	Acc: 88.46%	Val Loss: 0.8805	Acc: 72.72%
Epoch 02/25	Train Loss: 0.2637	Acc: 98.03%	Val Loss: 0.2566	Acc: 98.33%
Epoch 03/25	Train Loss: 0.2264	Acc: 99.42%	Val Loss: 0.2223	Acc: 99.19%
Epoch 04/25	Train Loss: 0.2161	Acc: 99.67%	Val Loss: 0.3361	Acc: 94.42%
Epoch 05/25	Train Loss: 0.2136	Acc: 99.69%	Val Loss: 0.2679	Acc: 97.22%
Epoch 06/25	Train Loss: 0.2108	Acc: 99.77%	Val Loss: 0.2089	Acc: 99.71%
Epoch 07/25	Train Loss: 0.2095	Acc: 99.80%	Val Loss: 0.2952	Acc: 96.10%
Epoch 08/25	Train Loss: 0.2090	Acc: 99.80%	Val Loss: 0.2186	Acc: 99.37%
Epoch 09/25	Train Loss: 0.2076	Acc: 99.82%	Val Loss: 0.2044	Acc: 99.87%
Epoch 10/25	Train Loss: 0.2061	Acc: 99.88%	Val Loss: 0.3748	Acc: 93.82%
Epoch 11/25	Train Loss: 0.2057	Acc: 99.88%	Val Loss: 0.2031	Acc: 99.92%
Epoch 12/25	Train Loss: 0.2044	Acc: 99.93%	Val Loss: 0.2025	Acc: 99.95%
Epoch 13/25	Train Loss: 0.2044	Acc: 99.93%	Val Loss: 0.2091	Acc: 99.71%
Epoch 14/25	Train Loss: 0.2039	Acc: 99.95%	Val Loss: 0.2354	Acc: 98.56%
Epoch 15/25	Train Loss: 0.2032	Acc: 99.97%	Val Loss: 0.2016	Acc: 99.98%
Epoch 16/25	Train Loss: 0.2030	Acc: 99.98%	Val Loss: 0.2014	Acc: 99.99%
Epoch 17/25	Train Loss: 0.2026	Acc: 99.98%	Val Loss: 0.2021	Acc: 99.97%
Epoch 18/25	Train Loss: 0.2024	Acc: 100.00%	Val Loss: 0.2019	Acc: 99.96%
Epoch 19/25	Train Loss: 0.2024	Acc: 99.99%	Val Loss: 0.2014	Acc: 99.98%
Epoch 20/25	Train Loss: 0.2022	Acc: 100.00%	Val Loss: 0.2015	Acc: 99.98%
Epoch 21/25	Train Loss: 0.2023	Acc: 100.00%	Val Loss: 0.2013	Acc: 100.00%
Epoch 22/25	Train Loss: 0.2023	Acc: 100.00%	Val Loss: 0.2013	Acc: 99.99%
Epoch 23/25	Train Loss: 0.2022	Acc: 100.00%	Val Loss: 0.2013	Acc: 99.99%
Epoch 24/25	Train Loss: 0.2021	Acc: 100.00%	Val Loss: 0.2012	Acc: 100.00%
Epoch 25/25	Train Loss: 0.2021	Acc: 100.00%	Val Loss: 0.2012	Acc: 100.00%

DementiaCNN test accuracy: 99.99%

Fig. 4. Results from the DementiaCNN model being applied to the Dementia dataset

Fig. 4 highlights DementiaCNN having the highest test accuracy out of the 3 models at 99.99%. It trained for the full 25 epochs. The validation loss kept improving by very small amounts late in training, so the early stopping patience of 6 was never reached, and the best weights from epoch 24 were used for testing. This model used AdamW optimizer with a learning rate of 3e-4 and weight decay of 1e-4. The learning rate had a smaller lr than LeNet, meaning it took smaller weight update steps. This was combined with the CosineAnnealingLR scheduler, which decreased the lr throughout training. A label smoothing of .05 was used to prevent overconfidence and gradient clipping to prevent instability. These techniques are the reason that DementiaCNN outperformed LeNet despite similar structure depth. This model has 196,388 trainable parameters, which is more than LeNet. One important statistic to note is that the model had a mean confidence of 0.963 for correct predictions and a mean confidence of .542 on its single incorrect prediction, meaning it was uncertain when wrong.

This dataset has a class imbalance caveat. As mentioned

earlier 77% of the dataset is from the Non Demented class. This means that a model predicting Non Demented every time would score a 77% without even knowing anything. So the 99% accuracy we saw was inflated by this. Moderate Dementia only has 488 total images, less than 1% of the dataset, leading to the model barely seeing Moderate Dementia during training. Precision, recall, and F1 per class are more honest metrics than overall accuracy in our case. Section C will reveal where the real failures are. This is one of the core limitations of evaluating on severely imbalanced medical datasets

C. Loss Curves, Accuracy Curves, and Confusion Matrices

This section presents training curves and classification scores for all 3 models being evaluated on the Dementia dataset.

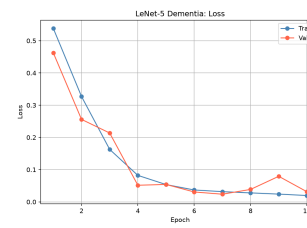


Fig. 5. Loss Curve from the LeNet model being applied to the Dementia dataset

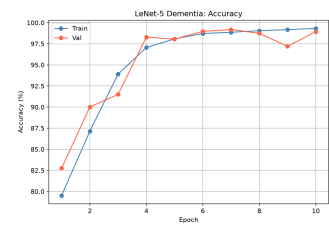


Fig. 6. Accuracy Curve from the LeNet model being applied to the Dementia dataset

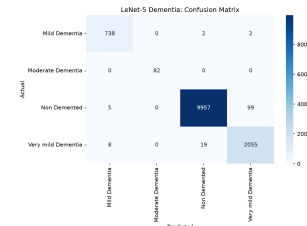


Fig. 7. Confusion Matrix from the LeNet model being applied to the Dementia dataset

	precision	recall	f1-score	support
Mild Dementia	0.9827	0.9946	0.9886	742
Moderate Dementia	1.0000	1.0000	1.0000	82
Non Demented	0.9979	0.9997	0.9998	10061
Very mild Dementia	0.9532	0.9870	0.9698	2082
accuracy	0.9834	0.9920	0.9886	12967
macro avg	0.9834	0.9920	0.9886	12967
weighted avg	0.9899	0.9896	0.9897	12967

Fig. 8. Classification Report from the LeNet model being applied to the Dementia dataset

As shown in Fig. 5 and Fig. 6, the loss decreased smoothly from 0.5382 to 0.0197. From epoch 4 onward the training and validation accuracies tracked within about 2% of each other, with only a brief validation dip at epoch 9. Both curves moved in the same direction, a clear sign of healthy learning. Overall, there are no signs of overfitting, meaning that LeNet learned genuine features of the MRI scans rather than memorizing training data. The smooth convergence suggests that the architecture was well suited for the dataset size and complexity.

LeNet's overall testing accuracy was 98.96% with a macro average F1 of .988 and a weighted average F1 of .990. We see in Fig. 8 the precision, recall, and F1 values were at .95 or higher across all classes, with Very Mild Dementia the weakest class at an F1 of .97. As illustrated in Fig. 7, the most common error was Non Demented scans being misclassified as Very Mild Dementia (99 images),

followed by Very Mild Dementia being misclassified as Non Demented (19 images). Both errors sit on the Non Demented and Very Mild boundary, supporting the low-saliency shape recognition argument. Even with this bit of confusion, the recall on minority classes is still very high, including perfect recall on Moderate Dementia. Having a macro F1 score of .988 is genuinely a strong result since macro average treats all classes as equal regardless of size.

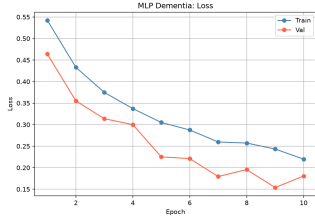


Fig. 9. Loss Curve from the MLP model being applied to the Dementia dataset

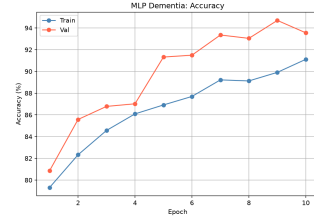


Fig. 10. Accuracy Curve from the MLP model being applied to the Dementia dataset

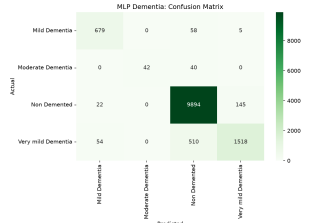


Fig. 11. Confusion Matrix from the MLP model being applied to the Dementia dataset

In the MLP training process, the loss decreased from .5420 to .2198, slower and less sharp than LeNet, as shown in Fig. 9. Having a final loss of .2198 indicates that the model did not converge well. As observed in Fig. 10 the validation accuracy was consistently higher than the training accuracy. This can be explained by dropout, which is active during training but switched off during validation. This also suggests the model did not overfit. The MLP model performed reasonably well despite its architectural limitations.

The overall test accuracy for the MLP model is 93.57% but as shown in Fig. 12 the macro average F1 is only .84 giving us a more honest score of the model. The weighted average is pulled up by a near perfect performance in the Non Demented class. As illustrated in Fig. 11, the model has clinically concerning recall scores on the other 3 classes. .92 on Mild Dementia, .51 on Moderate Dementia, and .73 on Very Mild Dementia. The model misses about 49% of Moderate Dementia cases, labeling 40 of the 82 test scans as Non Demented, which would mean roughly 1 in every 2 patients with moderate dementia being told they are fine. It also labels 510 Very Mild Dementia scans as Non Demented. Another finding is Moderate Dementia having a precision score of 1.00 but a recall score of .51, this means every time the model classifies an image

	precision	recall	f1-score	support
Mild Dementia	0.8993	0.9151	0.9071	742
Moderate Dementia	1.0000	0.5122	0.6774	82
Non Demented	0.9421	0.9834	0.9623	10061
Very mild Dementia	0.9101	0.7291	0.8096	2082
accuracy	0.9379	0.7849	0.8391	12967
macro avg	0.9349	0.9357	0.9328	12967

Fig. 12. Classification Report from the MLP model being applied to the Dementia dataset

as Moderate Dementia, it is correct, but the model is conservative when classifying as Dementia, causing the 49% error rate mentioned above.

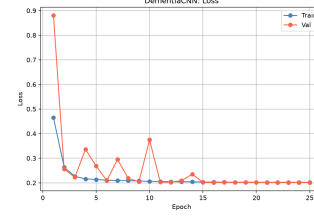


Fig. 13. Loss Curve from the DementiaCNN model being applied to the Dementia dataset

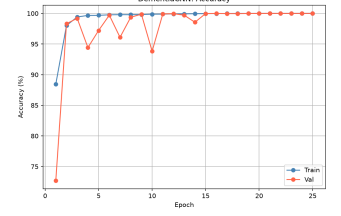


Fig. 14. Accuracy Curve from the DementiaCNN model being applied to the Dementia dataset

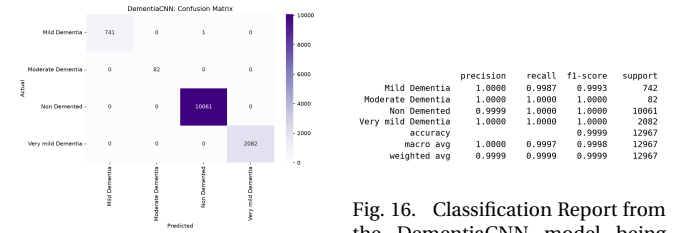


Fig. 15. Confusion Matrix from the DementiaCNN model being applied to the Dementia dataset

Fig. 16. Classification Report from the DementiaCNN model being applied to the Dementia dataset

	precision	recall	f1-score	support
Mild Dementia	1.0000	0.9987	0.9993	742
Moderate Dementia	1.0000	1.0000	1.0000	82
Non Demented	0.9999	1.0000	1.0000	10061
Very mild Dementia	1.0000	1.0000	1.0000	2082
accuracy	0.9999	0.9997	0.9998	12967
macro avg	1.0000	0.9997	0.9998	12967
weighted avg	0.9999	0.9999	0.9999	12967

In the DementiaCNN training process, the loss decreased from .4655 to .2021 across all 25 epochs, as shown in Fig. 13. Validation loss spiked at epochs 1, 4, 7 and 10, dropping the validation accuracy to 72.72% at epoch 1, 94.42% at epoch 4, 96.10% at epoch 7 and 93.82% at epoch 10. This early instability suggests the SE blocks and depthwise separable convolutions needed more epochs to learn meaningful attention weights. Despite this early instability the model recovered and stabilized after epoch 10. This can be attributed to gradient clipping and cosine scheduling, without them, the model may have diverged entirely. The best model was saved at epoch 24 with a validation loss of .2012 and a validation accuracy of 100%, and because the validation loss kept improving by tiny amounts, early stopping was never triggered.

DementiaCNN's overall test accuracy is 99.99%, the highest of all 3 models, as illustrated in Fig. 14. The macro average F1 is also at .9998, meaning the model performed equally well on classes regardless of their size. As shown in Fig. 15, the model had perfect recall on Moderate Dementia, a class with only 82 test samples and only 488 images in total. This is very impressive and shows that SE attention blocks genuinely learned important features for this class rather than relying on frequency. The model made a single error on the whole test set, a Mild Dementia scan misclassified as Non Demented. That error sits on the same boundary between early stage dementia and Non Demented scans that caused most of the mistakes of the other models, which is consistent with the low-saliency argument. Comparing DementiaCNN's results to

the other 2 models, as seen in Fig. 16, shows advanced training techniques such as SE blocks, label smoothing, cosine scheduler, and early stopping make a measurable difference on imbalanced medical imaging tasks.

All three models had the most trouble separating Non Demented from Very Mild and Mild Dementia, while Non Demented was the strongest class. Moderate Dementia was the most unreliable class for the MLP due to its underrepresentation, although LeNet still reached perfect recall on it. MLP and LeNet both showed consistent confusion between Mild, Very Mild, and Non Demented images. DementiaCNN did not follow this pattern and had near perfect results. The MLP mostly erred in the direction of Non Demented, missing dementia cases rather than raising false alarms, while LeNet's most frequent error went the other way and labeled Non Demented scans as Very Mild Dementia. In both cases the mistakes concentrate on the least salient boundary, which further supports the low-salience argument. DementiaCNN mostly breaking this pattern through SE attention suggests feature selection is more important than raw architecture depth in this use case.

Using overall test accuracy as a metric on this dataset can be misleading due to class imbalance. Instead, macro average F1 would be the more honest metric on this dataset. LeNet had a macro F1 of .988, MLP had .839, and DementiaCNN had .9998. MLP having a 9 point gap between weighted (.933) and macro F1s quantifies the class imbalance effect. For DementiaCNN and LeNet the gap is negligible, meaning they both performed very well on the underrepresented classes. The overall ranking of models would be DementiaCNN, LeNet, MLP from best to worst.

Having Non Demented be 77% of the dataset inflates the accuracy numbers. Moderate Dementia only takes 1% of the dataset, meaning the model has not seen it as much as it should. Recall is the most important metric because missing a real dementia case would have real life consequences. MLP and LeNet have room for improvement on the underrepresented classes, DementiaCNN is already near perfect.

D. Fashion MNIST results

Both LeNet and MLP were evaluated on Fashion MNIST. LeNet had a 89.45% test accuracy while MLP had a 86.91% test accuracy

As shown in Fig. 17 and Fig. 18, the LeNet 5 model was able to achieve a 89.45% test accuracy, which was about what was expected. The training accuracy was 92.61% and the validation accuracy was 89.62% after epoch 10. The loss constantly decreased throughout the epochs, starting at .5816 and ending at .1988. The results of the training and validation accuracies tracked within 3% of each other and moved in the same direction throughout training, meaning there are no signs of significant overfitting. The strongest performance came in structured categories such

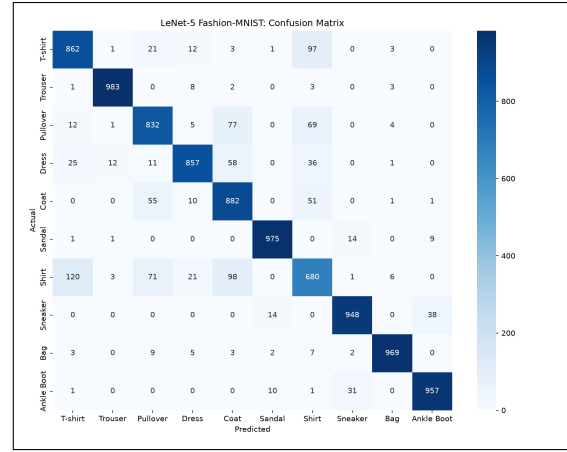


Fig. 17. Confusion Matrix from the LeNet model being applied to the Fashion MNIST dataset

	precision	recall	f1-score	support
T-shirt	0.8410	0.8620	0.8514	1000
Trouser	0.9820	0.9830	0.9825	1000
Pullover	0.8328	0.8320	0.8324	1000
Dress	0.9336	0.8570	0.8936	1000
Coat	0.7854	0.8820	0.8309	1000
Sandal	0.9731	0.9750	0.9740	1000
Shirt	0.7203	0.6800	0.6996	1000
Sneaker	0.9518	0.9480	0.9499	1000
Bag	0.9818	0.9690	0.9753	1000
Ankle Boot	0.9522	0.9570	0.9546	1000
accuracy			0.8945	10000
macro avg	0.8954	0.8945	0.8944	10000
weighted avg	0.8954	0.8945	0.8944	10000

Fig. 18. Classification Report from the LeNet model being applied to the Fashion MNIST dataset

as Trouser and Bag. The model did struggle on visually similar categories like Shirt, T-Shirt, and Pullover.

As shown in Fig. 19 and Fig. 20, MLP showed a test accuracy of 86.91%, which is, as expected, lower than LeNet 5's accuracy. The training accuracy was 87.32% and the validation accuracy was 88.10% after epoch 10. The loss also constantly decreased, starting at .5964 and ending at .3476. MLP flattens the image to a 1024 dimensional vector, where it loses all spatial relationships. The MLP model also has no convolutional layers, which means there is no edge or texture detection. Due to this the lower results were expected and overall make sense. An 87% accuracy is not a bad result on this dataset and shows the pixel level patterns alone carry some signal. The weakest performance for this model was with the Shirt, Coat, and Pullover categories.

LeNet outperformed MLP by about 2.5% overall. This gap confirms that spatial feature extraction does add measurable value on image data. This also confirms CNNs are better suited for image classification. Both models show consistent training curves, leading us to believe there are no major signs of overfitting. It can be assumed that LeNet benefited from its convolutional layers detecting edges and textures in clothing.

Both models struggled with the same classes with the main ones being Shirt, Coat, T-shirt, Pullover. This struggle

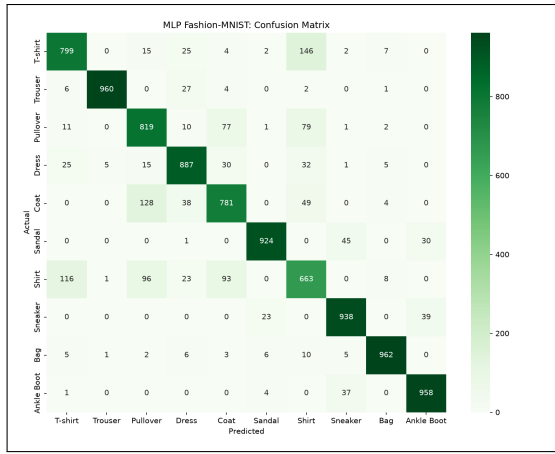


Fig. 19. Confusion Matrix from the MLP model being applied to the Fashion MNIST dataset

	precision	recall	f1-score	support
T-shirt	0.8297	0.7990	0.8141	1000
Trouser	0.9928	0.9600	0.9761	1000
Pullover	0.7619	0.8190	0.7894	1000
Dress	0.8722	0.8870	0.8795	1000
Coat	0.7873	0.7810	0.7841	1000
Sandal	0.9625	0.9240	0.9429	1000
Shirt	0.6758	0.6630	0.6694	1000
Sneaker	0.9116	0.9380	0.9246	1000
Bag	0.9727	0.9620	0.9673	1000
Ankle Boot	0.9328	0.9580	0.9452	1000
accuracy			0.8691	10000
macro avg	0.8699	0.8691	0.8693	10000
weighted avg	0.8699	0.8691	0.8693	10000

Fig. 20. Classification Report from the MLP model being applied to the Fashion MNIST dataset

can be attributed to the visual similarity of the classes, not the architecture itself. Since both models struggled on those classes, it can be validated that those classes are overall difficult to distinguish between. On the other hand, the Trouser and Bag categories had the best results, probably due to their visual distinction from other categories. There was some confusion between Sandals and Sneakers that can be attributed to their similar structure.

The accuracy on Fashion MNIST seems significantly worse than the accuracy on the Dementia dataset. This seems counterintuitive at first because Fashion MNIST should be the easier dataset, but the reason for this is how well balanced Fashion MNIST is compared to the Dementia Dataset. Fashion MNIST is actually a harder and better benchmark due to its balance.

E. Dataset Visualization

As evident from examining the number of images per class, clusters are dominated by the non-demented class (Green clusters), which accounts for up to 77 percent of the total dataset.

We decided to test visualization with different degrees of n -neighbors, and what we found was interesting. Moderate Dementia clusters often overlapped heavily with other classes. This got worse as we increased the number of neighbors each cluster can have. The class itself is

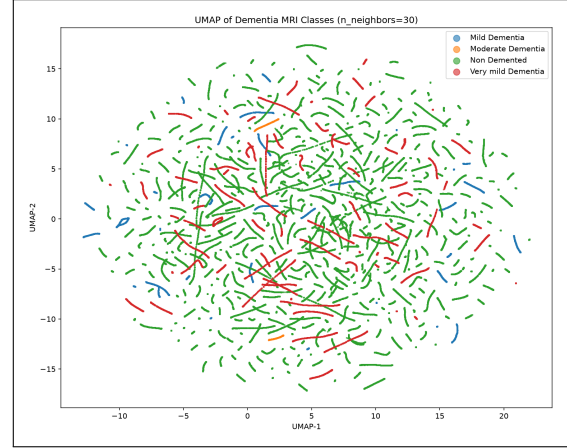


Fig. 21. UMAP Data Label Visualization (n=30)

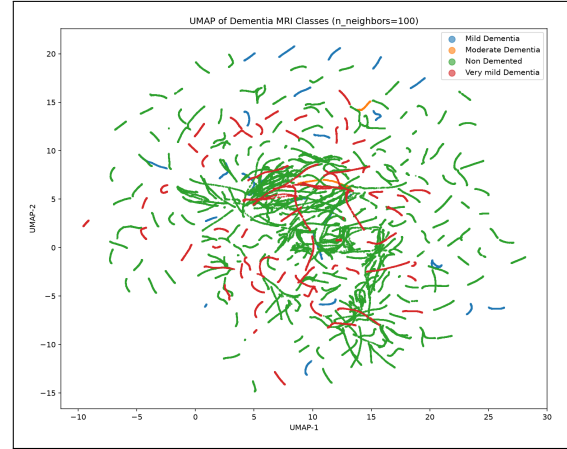


Fig. 22. UMAP Data Label Visualization (n=100)

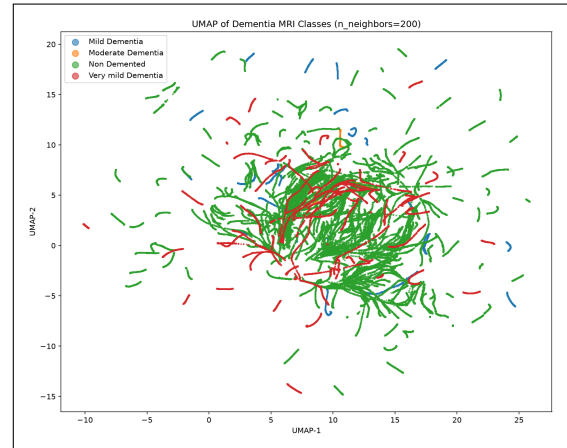


Fig. 23. UMAP Data Label Visualization (n=200)

practically invisible when $n = 200$. Mild dementia clusters are spaced out with some overlap, while very mild is typically central. With this, our prediction was that the models would have the biggest issue classifying Moderate and Mild dementia scans, and while this prediction held up somewhat with the MLP model(it was very inaccurate overall), this wasn't the case for LeNet and DementiaCNN.

MLP f1-scores showed that the model performed great on non-demented instances, especially on identifying the amount of positive non-dementia cases there were, but performed poorly (0.51) on recalling Moderate Dementia. The precision on cases of dementia of any scale was a lot higher in comparison to its recall, which tells us that the model was hesitant in identifying cases of any form of dementia.

Due to the fact that MLP flattens its images before ever processing them, identifying atrophy(shrinkage on the outer layer of the brain) becomes very difficult as you lose information on thickness, volume, and pixel relationship that are essential in cases of cortical thinning, hippocampal shrinkage, and asymmetry between hemispheres.

Apart from obvious performance improvements, We did not know what to expect from LeNet-5 and DementiaCNN from this cluster arrangement, but it was surprising to see that LeNet-5 still struggles to some degree classifying Very Mild cases of dementia as opposed to Non Demented, which means the dynamic and selective feature extraction capabilities of our custom model specifically made classification improvements by deciphering non-important edges. In the case of Very Mild vs Non demented this would be very important as atrophy isn't very pronounced

F. Edge Sensitivity Analysis

1) *LeNet-5 Results:* Edge analysis was conducted in this project to investigate whether image boundary complexity relates to classification accuracy. Three metrics were used, Canny edge detection, Sobel gradient magnitude, and a custom edge irregularity score. Canny detects binary edges, Sobel measures continuous gradient strength and the custom edge irregularity combines edge density with directional variance. The edge analysis was conducted based on the LeNet model and included its results in both Fashion MNIST and Dementia dataset. It was established from the Bhalla et al. [8] paper that edge information is a meaningful metric for image quality and model performance. This paper introduces the irregularity metric as a more nuanced measure of boundary complexity.

As shown in Fig. 24, the images with the lowest edge scores, below about 32, had the highest accuracies at around 90-100%. Once edge sharpness went over about 32, the accuracy declined fairly steadily, reaching its lowest point of around 72% at scores of 76-80, before a small recovery in the highest bins where only a few images fall. This suggests that images with many edges introduced visual complexity that confused the model. This would make sense because a plain t-shirt with clean edges would

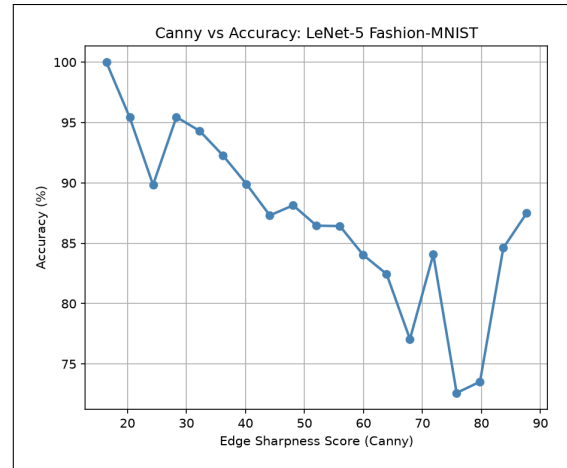


Fig. 24. Canny score vs Accuracy on LeNet's results on Fashion MNIST

be easier to classify than a heavily textured coat with many competing edges. Incorrectly classified images had an average Canny score of 47.43 while correctly classified ones had an average Canny score of 42.18.

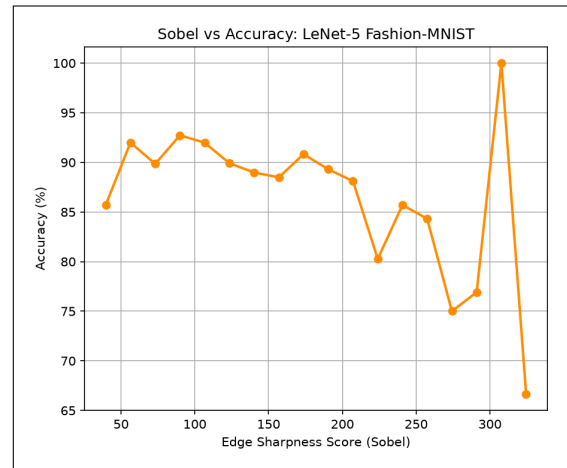


Fig. 25. Sobel score vs Accuracy on LeNet's results on Fashion MNIST

As observed in Fig. 25, low Sobel scores around 40 had an accuracy of around 86%. Accuracy peaked at about 88-93% for scores between roughly 55 and 210. Above about 210 the accuracy fell to between 75% and 86%, and the highest bins jumped between 67% and 100%, which is too small of a sample size to produce any meaningful conclusions. Overall, Sobel revealed a curve rather than a straight line. Too little and too much gradient strength would hurt accuracy. The range between 55 and 210 represents images with just enough gradient information for LeNet to extract meaningful spatial features without being overwhelmed by complexity. Incorrectly classified images average Sobel scores of 153.68 while correctly classified ones had an average score of 148.31.

As illustrated in Fig. 26, irregularity scores below .2 swung widely, from 100% in the smallest bins to around 83% at .14, because only a few images fall there. Between .2 and .45 the accuracy stayed between roughly 87% and

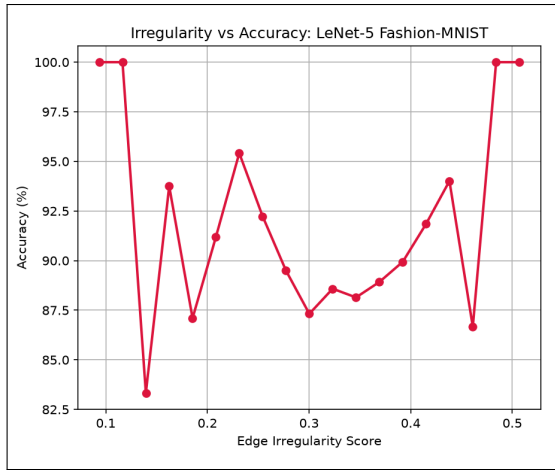


Fig. 26. Edge Irregularity score vs Accuracy on LeNet’s results on Fashion MNIST

95%, with the best results around .23, and the highest bins again reached 100% with very few images. Unlike Canny and Sobel, irregularity shows no single clear trend on Fashion MNIST, and the average irregularity of correct (.3184) and incorrect (.3204) predictions is very close. The edge irregularity metric captures wrinkles and directional chaos in edges, which in this dataset could be fabric texture and folds, but on its own it is a weaker predictor of difficulty than edge sharpness.

Canny and Sobel both showed that higher edge complexity tends to hurt the model, while irregularity was flatter. Incorrectly classified images had higher average scores than correctly classified images for all three metrics, with the clearest gap for Canny. The cluster of Shirt, Coat, T-shirt, and Pullover that caused the models to struggle the most probably had the most edge overlap. Similar textures and irregularity scores make them hard to distinguish. We can conclude that edge sharpness is a meaningful predictor of classification difficulty in the Fashion MNIST dataset. LeNet performs best on images with moderate and well defined boundaries, which is what LeNets convolutional filters were designed to process.

As shown in Fig. 27, the accuracy stayed between 97.3-100% throughout all edge score bins, which is an extremely narrow range. There was no clear trend, accuracy fluctuated randomly throughout edge scores. The average Canny scores for correct (67.18) and incorrect (66.26) predictions only have a .92 difference, which is negligible. This confirms that Canny edge sharpness has no relationship with the classification of Dementia MRIs. The model having nearly the same Canny scores for correct and incorrect is the opposite of Fashion MNIST where the scores had significant gaps.

As observed in Fig. 28, accuracy stayed between 98.5-100% in most Sobel score bins. There was an outlier drop to about 94% around scores of 67 and a smaller dip to 97.5% around 137. Outside these drops, the accuracy was nearly perfect in the rest of the bins. The average Sobel score for correct (103.22) and incorrect (101.93) predic-

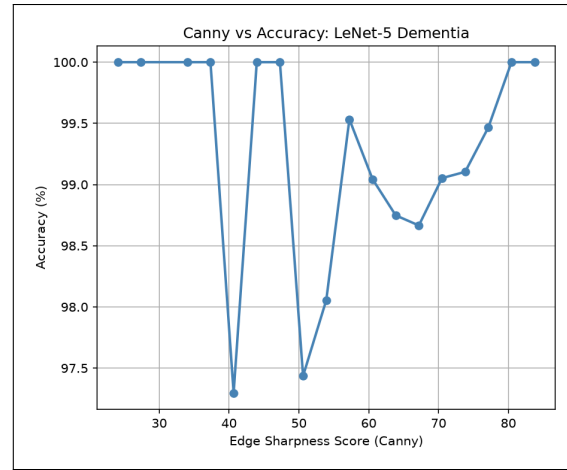


Fig. 27. Canny score vs Accuracy on LeNet’s results on the Dementia dataset

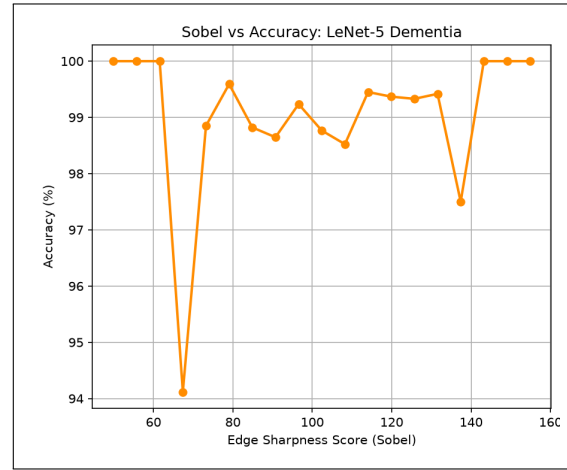


Fig. 28. Sobel score vs Accuracy on LeNet’s results on the Dementia dataset

tions only has a 1.29 difference, which again is negligible. This means that gradient strength has no meaningful relationship with whether the model classifies a dementia MRI correctly. The sharp outlier can be attributed to noise from a small bin rather than a pattern. This is again a contrast from Fashion MNIST, where accuracy clearly fell as gradient strength grew.

As illustrated in Fig. 29, accuracy stayed between 97-100% at almost every irregularity score bin. There were two sharp drops, to about 95.6% at .22 and about 96.4% at .36, but those can again be attributed to noise from small bins. The average irregularity score for correct (.2900) and incorrect (.2935) predictions only has a .0035 difference, again a negligible amount. This means that Edge irregularity also fails to predict classification of dementia MRIs. The y-axis only having about a 4% gap tells us the complexity of the edges barely changes accuracy.

2) *MLP Results:* The results of edge sensitivity analysis for the MLP model showed a U-shaped distribution of accuracy for all three edge scores (canny, sobel, irregularity), as shown in Fig. 30, Fig. 31 and Fig. 32. Accuracy was

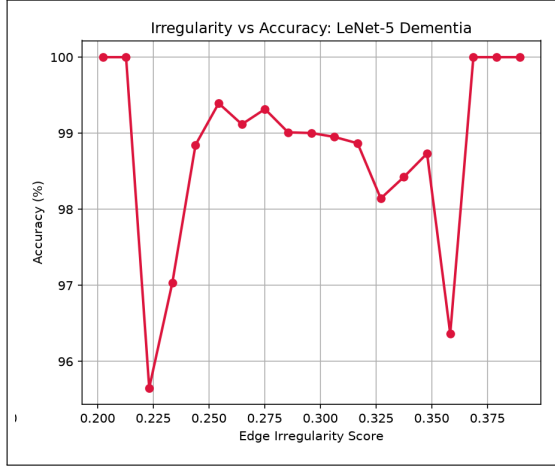


Fig. 29. Edge Irregularity score vs Accuracy on LeNet's results on the Dementia dataset

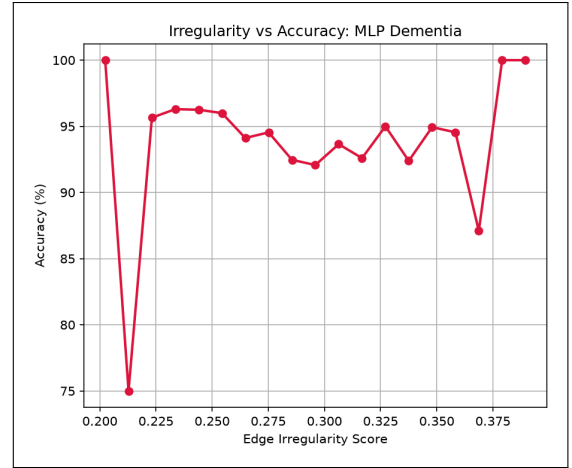


Fig. 32. Edge Irregularity vs Accuracy on MLP's results on the Dementia dataset

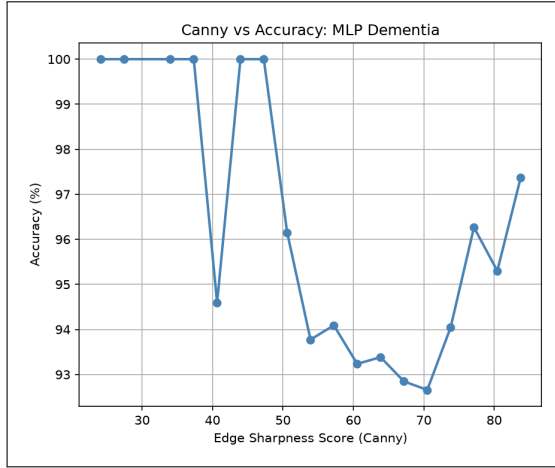


Fig. 30. Edge Canny Sharpness score vs Accuracy on MLP's results on the Dementia dataset

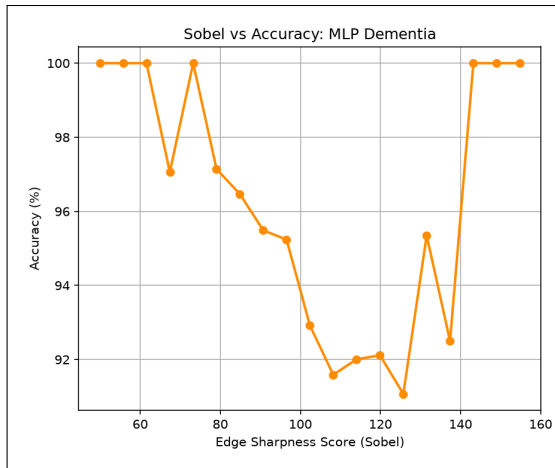


Fig. 31. Edge Sobel Sharpness score vs Accuracy on MLP's results on the Dementia dataset

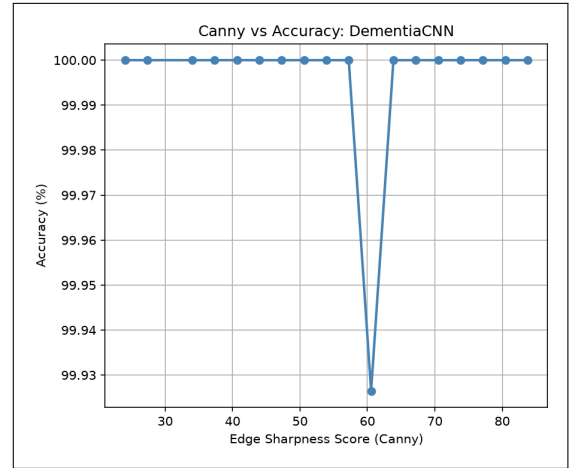


Fig. 33. Edge Canny Sharpness score vs Accuracy on DCNN's results on the Dementia dataset

highest at the lowest edge scores, dropped to its lowest values in the mid-range (where the four dementia classes are most visually ambiguous), and recovered at the highest edge scores (where the scans are more visually distinct). Note that for these plots the range of accuracy is wider on the y-axis, at about 91-100% for Canny and Sobel, with one small irregularity bin dropping to 75%. The average scores of correct and incorrect predictions were again very close, 67.17 against 67.17 for Canny, 103.01 against 106.08 for Sobel and .2899 against .2921 for irregularity. Interestingly, the MLP has no spatial information whatsoever, as it processes an image by flattening each one to a 1024 dimensional vector. Although the two types of models are different in architecture, the edge profile sensitivities of both models are similar, indicating that the common limitation is not spatial processing, but rather the lack of any mechanism for selectively attending to or reweighting features according to their diagnostic value.

3) *DementiaCNN Results*: The accuracy distribution of the DementiaCNN model was very different from LeNet and MLP for all three edge metrics, as shown in Fig. 33,

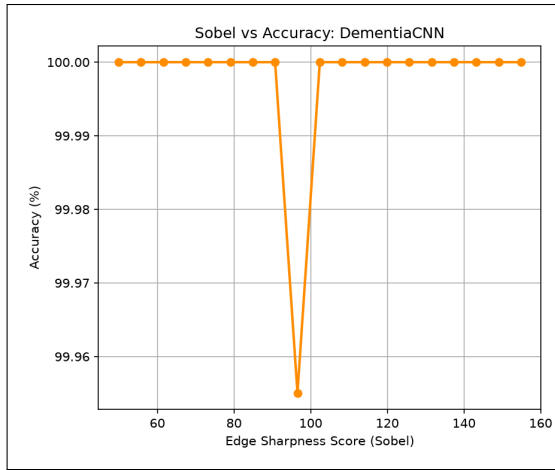


Fig. 34. Edge Sobel Sharpness score vs Accuracy on DCNN's results on the Dementia dataset

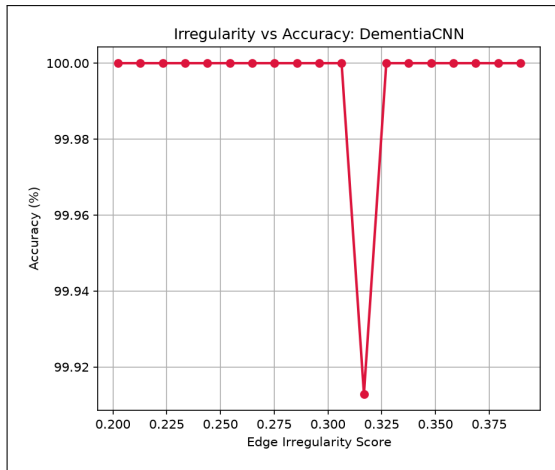


Fig. 35. Edge Irregularity vs Accuracy on DCNN's results on the Dementia dataset

Fig. 34 and Fig. 35. Accuracy was 100% in every Canny, Sobel and irregularity bin except one, and that single dip to about 99.9% comes from the one misclassified test image. This is a model whose decision boundary is not tied to any surface level edge property of the image, but has evolved a stable decision boundary, although the patient-level split limitation discussed in Section G applies to this result as well.

The architecture contains a few elements that are lacking in both LeNet and MLP, which together account for this behavior. Specifically, there are three convolutional blocks followed by most of the Squeeze-and-Excite blocks. The SE block works by first taking spatial features (the same as the other blocks), then summarizing information across all channels in each block, sending that summary through a small learned network, and multiplying the output of that network by the feature maps, per channel. This enables the model to adaptively filter out noise and/or uninformative texture from input signals, and boost the signal from input channels with diagnostic information, such as faint cortical thickness or sharp ventricular bor-

ders. Of course, this reweighting occurs on a per-image basis, so the reweighting is not a fixed importance ranking, but rather it's adapting on a per-image basis.

In addition to the SE blocks, DementiaCNN also introduces Batch Normalisation at all convolutional layers to normalise the activations while lessening the model's sensitivity to changes in input contrast and brightness that directly impact the distributions of edge scores. A depthwise-separable convolution operation further enhances the spatial detail following the third block, but without a further spatial downsampling, retaining fine-grained details before collapsing the representation to a single vector per channel using the Global Average Pooling head. The classifier head also performs dropout at two points and it was trained with label smoothing which helps to decrease overconfidence in the case of an ambiguous output. Together with a cosine annealing learning rate schedule (25 epochs as opposed to 10 epochs for LeNet and MLP), these design choices yielded a model whose accuracy is significantly less dependent upon an input image's high or low scores for raw edge metrics, which is exactly what the flat accuracy across every edge bin represents.

All three metrics reached the same conclusion, edge complexity does not predict classification accuracy on Dementia MRIs. The average score differences between correct and incorrect predictions are negligible for LeNet and MLP on all 3 metrics, and DementiaCNN made only one error, so its averages carry no real signal. The y-axis is very narrow, mostly sitting between 91-100%, while the Fashion MNIST y-axis goes from about 67-100% and shows a clear downward trend. Dementia MRI misclassifications are not explained by image quality, edge sharpness, or boundary complexity. These errors instead happen because of the biological differences between the stages of dementia. Key features such as subtle brain atrophy and minor tissue changes do not manifest themselves as detectable edge patterns at 32x32 resolution. The model is not failing because the images are visually poor, it is failing because the distinguishing information between classes is not visible at the edge level. This is the strongest evidence to support the low-salience shape recognition hypothesis introduced in the introduction

Overall, Fashion MNIST showed a clear relationship between edge complexity and accuracy, while Dementia MRI did not. The classes of Fashion MNIST differ visually, an example being trousers having different edge patterns than a bag. In Dementia MRI the classes differ biologically in ways that do not show up in edge measurements. A Mild Dementia and Non Demented MRI scan have nearly identical edge profiles. This is a perfect example of low-salience with the distinguishing features not being visually sharp or well defined.

This edge analysis provides quantitative evidence for the low-salience shape recognition hypothesis from the introduction. Fashion MNIST is a high salience dataset with edge-accuracy correlation. Dementia MRI is a low-

salience dataset with no edge-accuracy correlation. This absence is not a failure of edge analysis, it is actually a finding that tells us that the information needed to distinguish dementia stages is not captured by low-level visual features like edges. Methods such as higher resolution imaging, 3D volumetric analysis, and richer feature extraction would be needed to capture the subtle biological distinctions between dementia stages that low-level edge features cannot detect.

G. Cross Dataset Comparison and Limitations

This section compares model performance across both datasets and identifies the key limitations of the study. The two datasets used represent opposite ends of the classification spectrum. Fashion MNIST is clean, balanced, and high-salience while dementia MRI is medical, imbalanced, and low-salience. The comparison is not only about which model performed best, it is also about understanding the conditions that make image classification hard and what architectural choices help overcome those conditions.

We discussed the results of the Dementia dataset in sections B and C and the results of Fashion MNIST in section D. We found that all models scored higher in raw accuracy in the Dementia dataset. This is not evidence that Dementia MRIs are easier to classify, it is instead evidence of how class imbalance inflates accuracy numbers. Fashion MNIST is a more honest benchmark at 90% accuracy than the Dementia dataset at 99% due to its class balance. LeNet outperformed MLP on both datasets, by about 2.5% on Fashion MNIST and about 5% on the Dementia dataset. This gap proves that spatial feature extraction adds value regardless of domain.

The gap between LeNet and MLP on Fashion MNIST does represent an architectural advantage. In the Dementia dataset, the overall accuracy gap is small but the macro F1 gap is large, .988 v .839. Macro F1 is the only metric that shows an honest cross dataset comparison. Fashion MNIST had lower macro F1 scores compared to the Dementia Dataset. Fashion MNIST was able to expose the true limits of both the MLP and LeNet models, being a better benchmark than its reputation suggested. Dementia CNN's improvement over LeNet on the Dementia dataset was small in overall accuracy but significant in macro F1 and underrepresented class recall.

The first limitation in our paper is the class imbalance in the Dementia dataset. The Non Demented class makes up 77% of the data set while Moderate Dementia makes up less than 1%. This imbalance inflates accuracy and makes the recall in underrepresented classes unreliable. We saw an example with MLP having a recall of .51 in Moderate Dementia. The other 2 models were still able to get high scores on it, but the limited amount of test data was not ideal. This limitation was the biggest limitation in this project.

The second limitation was the resolution of the MRI images. All models were trained on 32x32 pixel images.

MRI scans contain more detail the higher their resolution is. Some of the subtle structural brain changes that distinguish dementia stages like atrophy patterns and tissue density changes may not be visible at 32x32. The edge analysis confirmed that distinguishing features do not manifest at the edge level at this resolution while the UMAP visualization confirmed classes are not separable in feature space at this resolution. Both of these findings point to resolution as a fundamental limitation. Higher resolution inputs such as 128x128 or 256x256 combined with deeper architectures would likely reveal features that were not shown in our resolution. This was the most important fundamental limitation of this project. This again connects to the low salience shape recognition argument.

The MLP was included as a lower bound baseline and was expected to perform worse than the other 2 models. Fully connected networks treat each pixel independently and lack the ability to capture spatial relationships [4]. This means the MLP model is not able to keep spatial structure well, so it loses a lot of the subtle structural irregularities in brain tissue [6], which can cause confusion. MLP still provided value to this paper as a lower bound baseline.

The third limitation is not using data augmentation. Data augmentation artificially increases dataset diversity and could have been used on the Dementia dataset to better generalize on underrepresented classes. The Kshatri and Singh paper [6] identified data augmentation as a solution to limited medical data. This would have greatly benefitted the Moderate Dementia class that only had 488 images. The original reason for not including data augmentation is Moderate Dementia was found to be very easy to classify by Dardouri [3] Having data augmentation would have significantly improved class recall for Moderate Dementia and should be implemented in any future work on this dataset.

Another important limitation is that the data was split by image, not by patient. Neighboring slices from the same scan very likely appear in both the training and test sets, so the high test accuracy of all three models probably overstates how well they would perform on new patients. A patient-level split is the most important change for future work.

The final limitation is the difference in training setups. LeNet and MLP used an Adam optimizer, fixed learning rate, 10 epochs, and a batch size 32. On the other hand, DementiaCNN used AdamW, a cosine scheduler, label smoothing, gradient clipping, early stopping, and 25 epochs. These differences make it difficult to isolate whether DementiaCNN had better results due to the architecture or training setup. It is difficult to say how much improvement comes from SE blocks compared to label smoothing or more epochs. This can be figured out in the future by running a controlled study where each one technique is removed at a time to isolate each technique's contributions.

Considering all limitations described above, several directions for future work emerge. The most pressing direction for future work is re-splitting the data by patient, so that no scan contributes slices to both the training and test sets. The next priority is addressing class imbalance using oversampling, weighted loss functions, or SMOTE. Increasing the resolution of the MRIs from 32x32 to 128x128 or 256x256 would reveal the subtle biological features that are invisible at the current resolution. Applying data augmentation on the Moderate Dementia class would artificially expand the number of images in the class which would directly contribute to better recall scores. A controlled ablation study should be run on DementiaCNN to isolate each contribution and provide clearer architectural insights. Future work should also investigate optimizing the best performing model for mobile deployment through techniques such as quantization and knowledge distillation. As mentioned in the Introduction, this project was motivated to provide accessible dementia screening tools for resource limited regions that lack clinical infrastructure. Addressing the limitations in this section represents a clear path toward a clinically meaningful automated screening tool.

VI. CONCLUSIONS (JOSHUA: 20%, GERARDO: 80%)

This paper evaluated three different neural network architectures, LeNet 5, MLP, and DementiaCNN. LeNet was the controlled baseline, MLP was the straw man baseline that confirmed the importance of spatial features, and DementiaCNN was a custom architecture demonstrating advanced training techniques make a measurable difference on medical data classification. The results demonstrate that texture and spatial feature recognition are the most critical factors for dementia stage classification. However, proper dynamic edge prioritization through techniques such as Squeeze-and-Excite attention blocks further improves classification performance, particularly in cases where the visual boundary between two classes is ambiguous. Edge irregularity was introduced as a novel evaluation metric combining edge density and directional variance. Classification results, UMAP visualization, and edge analysis all converged to support the low-saliency shape recognition hypothesis introduced in the Introduction. This paper identified class imbalance and resolution as its primary limitations and proposes remedies to be implemented in future work. This work represents a step toward a clinically meaningful automated dementia screening tool accessible to resource-limited communities that lack the infrastructure for traditional diagnostic methods.

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REFERENCES

- [1] A. Mollalo, M. Kramer, M. Cutty, and B. Hoseini, "Systematic review and meta-analysis of rural-urban disparities in alzheimer's disease dementia prevalence," *The Journal of Prevention of Alzheimer's Disease*, p. 100305, 2025. [Online]. Available: <https://doi.org/10.1016/j.tpad.2025.100305>
- [2] M. Rahman, E. M. White, C. Mills, K. S. Thomas, and E. Jutkowitz, "Rural-urban differences in diagnostic incidence and prevalence of alzheimer's disease and related dementias," *Alzheimer's & Dementia*, vol. 17, no. 7, 2021. [Online]. Available: <https://doi.org/10.1002/alz.12285>
- [3] S. Dardouri, "An efficient method for early alzheimer's disease detection based on mri images using deep convolutional neural networks," *Frontiers in Artificial Intelligence*, vol. 8, 2025. [Online]. Available: <https://www.frontiersin.org/journals/artificial-intelligence/articles/10.3389/frai.2025.1563016/full>
- [4] Y. LeCun, L. Bottou, Y. Bengio, and P. Haffner, "Gradient-based learning applied to document recognition," *Proceedings of the IEEE*, vol. 86, no. 11, pp. 2278–2324, 1998. [Online]. Available: <http://yann.lecun.com/exdb/publis/pdf/lecun-01a.pdf>
- [5] H. Xiao, K. Rasul, and R. Vollgraf, "Fashion-mnist: a novel image dataset for benchmarking machine learning algorithms," *arXiv preprint arXiv:1708.07747*, 2017. [Online]. Available: <https://arxiv.org/abs/1708.07747>
- [6] S. S. Kshatri and D. Singh, "Convolutional neural network in medical image analysis: A review," *Archives of Computational Methods in Engineering*, vol. 30, pp. 2793–2810, 2023. [Online]. Available: <https://doi.org/10.1007/s11831-023-09898-w>
- [7] W. Lin, T. Tong, Q. Gao, D. Guo, X. Du, Y. Yang, G. Guo, M. Xiao, M. Du, and X. Qu, "Convolutional neural networks-based mri image analysis for the alzheimer's disease prediction from mild cognitive impairment," *Frontiers in Neuroscience*, vol. 12, p. 777, 2018. [Online]. Available: <https://doi.org/10.3389/fnins.2018.00777>
- [8] K. Bhalla, D. Koundal, B. Sharma, Y.-C. Hu, and A. Zaguia, "A fuzzy convolutional neural network for enhancing multi-focus image fusion," *Journal of Visual Communication and Image Representation*, vol. 84, p. 103485, 2022. [Online]. Available: <https://doi.org/10.1016/j.jvcir.2022.103485>